Kyma Companion

History & Current state

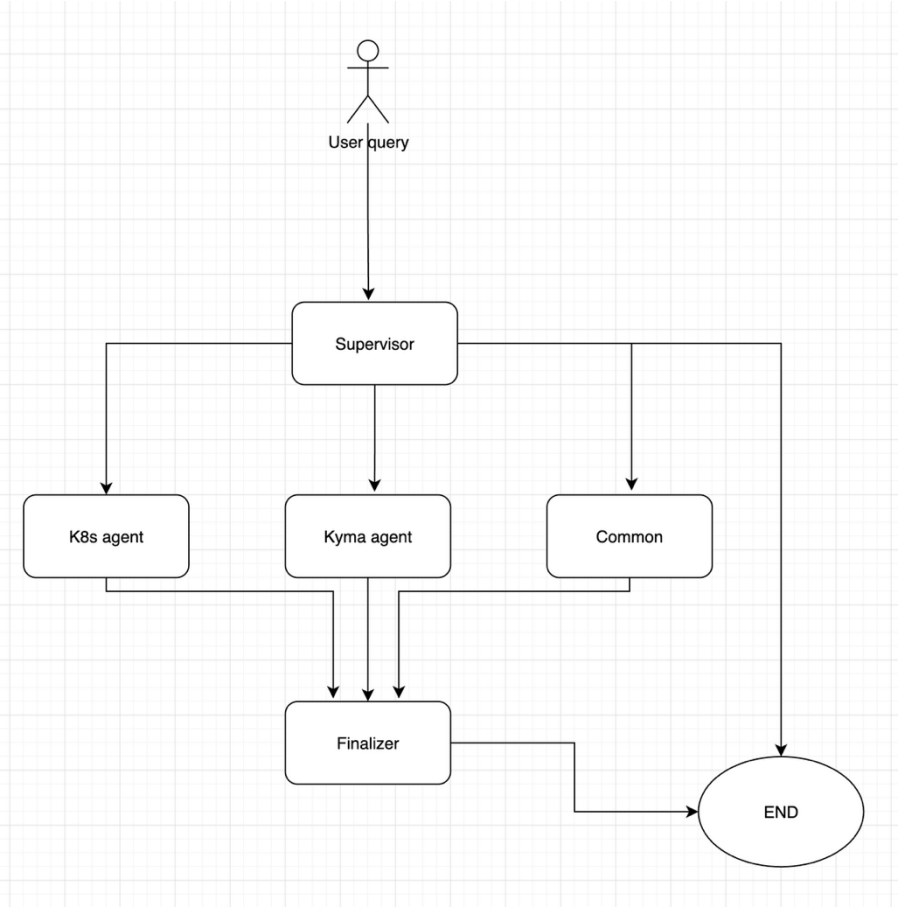
Hierarchical Multi-Agent System

April 2026

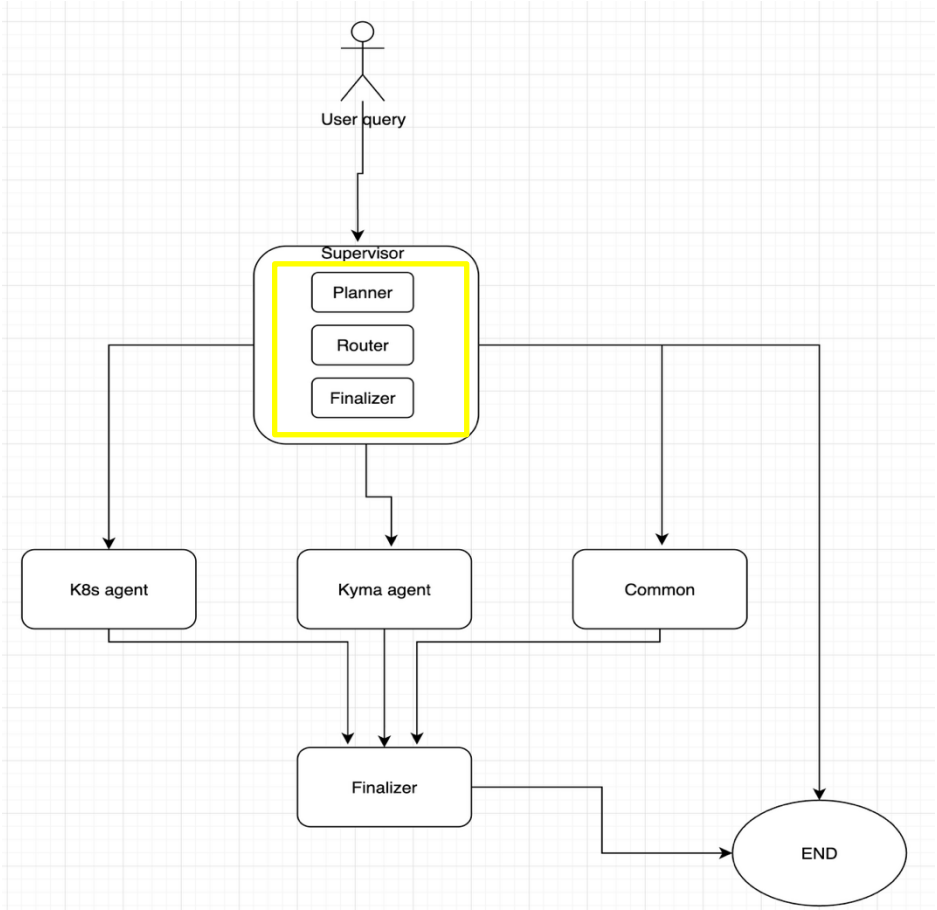
Architecture Overview

Hierarchical multi-agent system built with LangGraph

Iteration 1



Iteration 2



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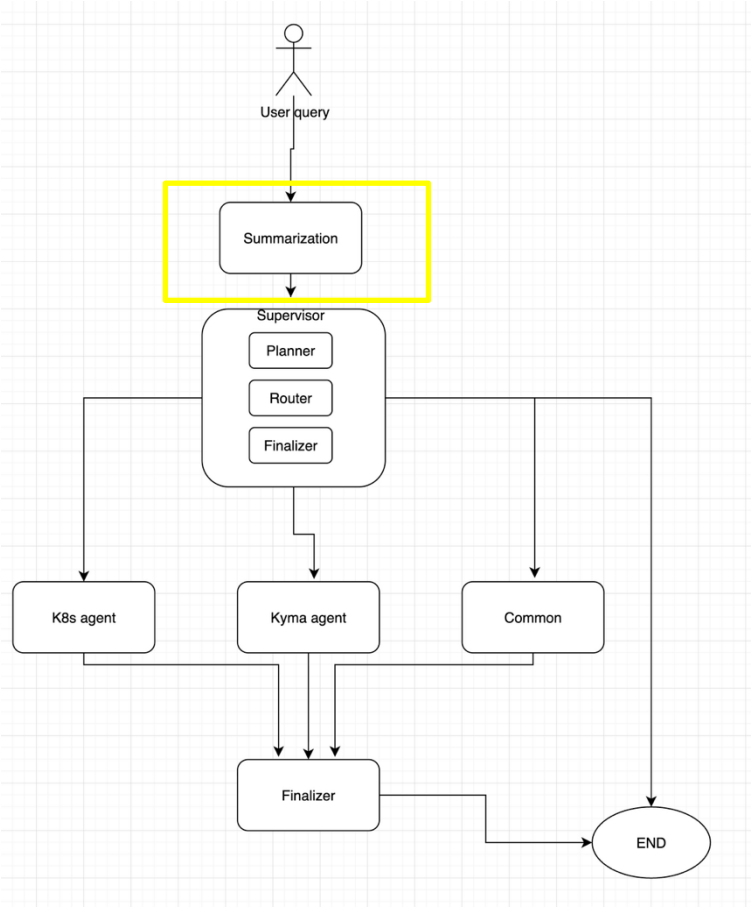
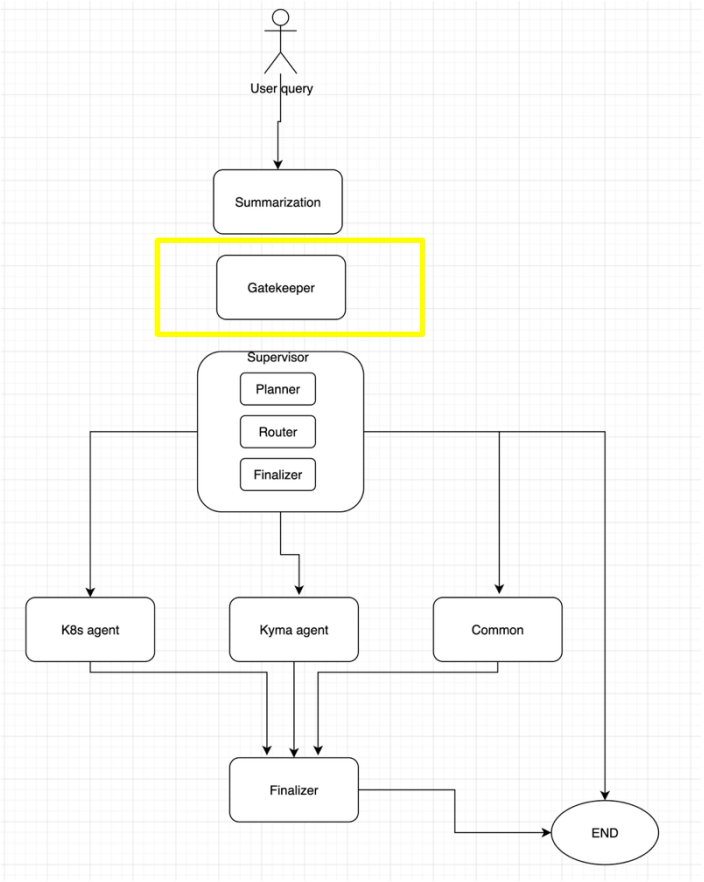
Iteration 3



Iteration 4



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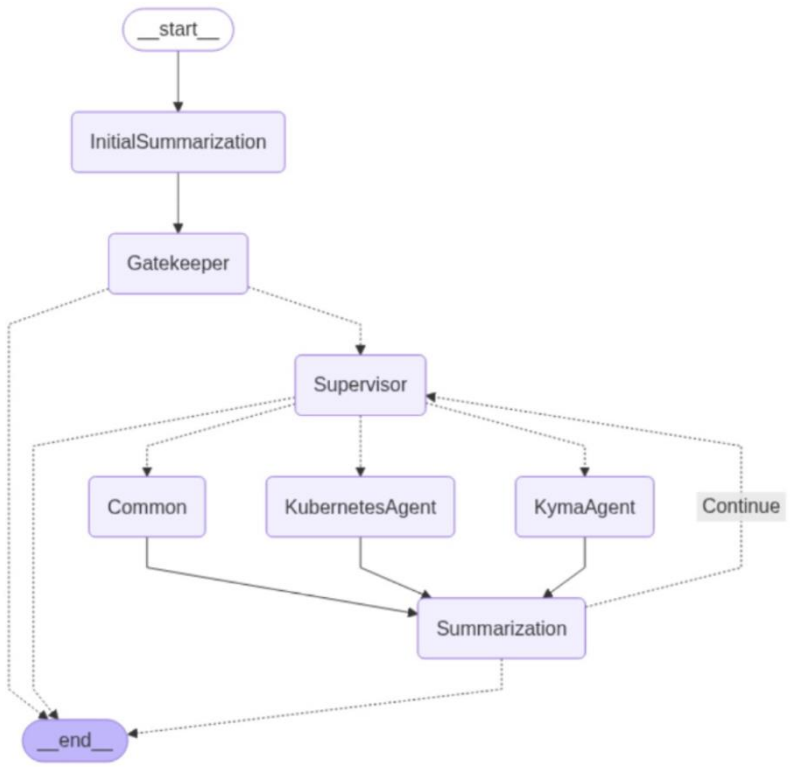
Current Graph



Query Lifecycle

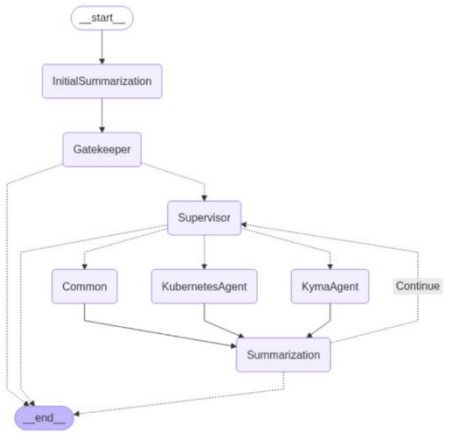
1	**Summarize**	Condense old message history. Condenses messages > 3k tokens on every request. Model: gpt-4o-mini
2	**Gatekeeper**	Security check + query classification + Direct response from message history. Model: gpt-4.1
3	**Supervisor**	Plan & dispatch to specialist agents. Orchestrates agents via Plan → Route → Finalize cycle
4	**Agents**	Kyma / Kubernetes / Common process tasks
5	**Finalizer**	Synthesize responses into final answer + filter inappropriate parts

Gatekeeper Node

Security gate and query classifier — Model: gpt-4.1

Responsibilities

- Detects prompt injection via system fields
- Detects security threats (RCE, SQLi, attack payloads)
- Classifies queries: Kyma, K8s, Programming, Greeting, Irrelevant
- Answers greetings directly (no forwarding)
- Answers Programming / About You queries via LLM
- Resolves past-tense Kyma/K8s queries from history
- Forwards present-tense Kyma/K8s queries to Supervisor
- Rejects out-of-domain queries with fixed message



Routing Rules

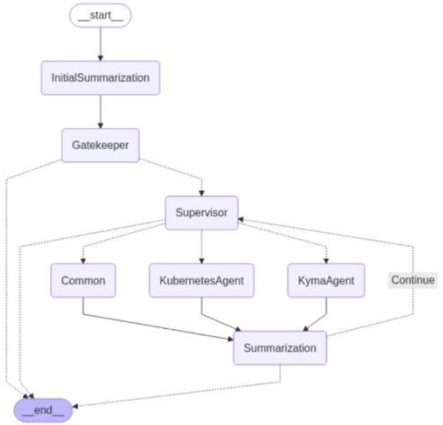
Prompt Injection detected	Block → outside-domain msg
Security threat detected	Block → outside-domain msg
Category: Irrelevant	Block → outside-domain msg
Category: Greeting	Fixed hello response
Kyma/K8s + past-tense	Answer from history
Kyma/K8s (present)	Forward → Supervisor
Programming / About You	Answer directly via LLM

Supervisor Subgraph

Plans, routes, and finalizes multi-agent responses

Planner

- Breaks query into ordered SubTask list
- Assigns to: KymaAgent, KubernetesAgent, Common
- Cluster-wide → both agents
- Namespace-wide → both (namespace-scoped)
- Mixed-domain → split by domain
- Ambiguous → defaults to KymaAgent
- General/programming → Common
- Model: gpt-4o-mini



Router

- Pure logic node — no LLM
- Scans subtasks for first PENDING item
- Sets 'next' = assigned agent name
- If no PENDING tasks → routes to Finalizer
- Determines which agent runs next

Finalizer

- Synthesizes all agent responses
- Never invents content beyond agent output
- Filters malicious content from responses
- Wraps new YAML in <YAML-NEW> tags
- Wraps update YAML in <YAML-UPDATE> tags
- All agents failed → standard failure msg
- Runs ResponseConverter post-processing
- Model: gpt-4.1

Agent Subgraph (KymaAgent & KubernetesAgent)

Shared architecture — each handles domain-specific subtasks

Execution Loop

subtask_selector	Find pending subtask
agent (LLM + tools)	Run LLM with bound tools
Summarization	Compress messages if needed
tools	Execute tool calls
(loop → agent)	Repeat until done
finalizer	Mark COMPLETED, clean up

Node Descriptions

subtask_selector

Scans state.subtasks for PENDING task assigned to this agent. If none found, routes to finalizer.

agent

Invokes LLM with tools on my_task.description + filtered history. Summarizes oversized tool responses. Stops early if remaining_steps ≤ 3.

Summarization

Agent-level token compression scoped to agent_messages / agent_messages_summary. Triggered when agent decides to call tools.

tools

LangGraph ToolNode executing tool calls from last AIMessage. handle_tool_errors=True — exceptions returned as ToolMessage for agent retry.

finalizer

Marks subtask COMPLETED (or keeps ERROR). Prepends subtask description for traceability. Promotes final result to parent messages list.

Agent Tools

Domain-specific tools available to each agent

KymaAgent Tools

fetch_kyma_resource_version

Discovers API group/version for Kyma resource kinds. Warns on deprecated versions. Uses K8s discovery endpoint.

kyma_query_tool

Executes raw GET against K8s API server via full URI. Returns Kyma resource JSON.

search_kyma_doc

RAG search over official Kyma docs. Returns top-5 relevant chunks (text-embedding-3-large). Filters by relevancy score ≥ 0.5.

KymaAgent Tool Flow

- Resource unknown: fetch_version → query → search_doc
- Resource known: query → search_doc
- On 404: query(err) → fetch_version → query(retry) → search_doc
- Conceptual queries: search_doc only

KubernetesAgent Tools

k8s_query_tool

Raw GET against K8s API server via full URI. Sanitizes sensitive data (e.g. removes Secret.data field).

fetch_pod_logs_tool

Fetches last 10 lines of current + previous logs for a pod container. Returns structured diagnostic context if current logs unavailable.

Common Node

No tools — answers directly from LLM knowledge and conversation history.
Used for: general programming questions, conceptual explanations, anything not requiring cluster access.
Model: gpt-4o-mini

Configuration & Guardrails

Limits that prevent runaway token usage, infinite loops, and context overload

Summarization & Step Limits

SUMMARIZATION_TOKEN_UPPER_LIMIT	3,000 tokens	Triggers summarization
SUMMARIZATION_TOKEN_LOWER_LIMIT	2,000 tokens	Target after trimming
TOOL_RESPONSE_TOKEN_COUNT_LIMIT	10,000 tokens	Summarize oversized tool responses
TOTAL_CHUNKS_LIMIT	2 chunks	Max chunks; else query rejected as too broad
remaining_steps (BaseAgentState)	25 steps	LangGraph-managed recursion counter
AGENT_STEPS_NUMBER	3 steps	Stop tool calls when ≤3 steps remain
GRAPH_STEP_TIMEOUT_SECONDS	180 s	Per-step wall-clock timeout for agents
RECENT_MESSAGES_LIMIT	10 messages	History window passed to Planner

API, RAG & Token Quota

K8S_API_PAGINATION_LIMIT	40 items	Items per page for K8s API listing
K8S_API_PAGINATION_MAX_PAGE	1 page	Only first page retrieved
POD_LOGS_TAIL_LINES_LIMIT	10 lines	Log lines per container
RAG_RELEVANCY_SCORE_THRESHOLD	0.5	Min cosine similarity for doc chunks
DEFAULT_TOP_K	5 chunks	Max doc chunks per search_kyma_doc call
TOKEN_LIMIT_PER_CLUSTER	5,000,000	Aggregate token cap per cluster (Redis-tracked)
TOKEN_USAGE_RESET_INTERVAL	86,400 s	Rolling 24h counter reset
MAX_TOKEN_LIMIT_INPUT_QUERY	8,000 tokens	Max tokens in single input query

Model Assignment

Each component uses a model appropriate to its complexity and cost requirements

Component → Model Mapping

Component	Model	Rationale
Gatekeeper	gpt-4.1	Security-critical; needs highest accuracy for injection/threat detection
KymaAgent	gpt-4.1	Complex Kyma domain reasoning + multi-step tool use
KubernetesAgent	gpt-4.1	Complex K8s domain reasoning + multi-step tool use
Finalizer	gpt-4.1	Full reasoning required to faithfully synthesize agent responses
Supervisor / Planner	gpt-4o-mini	Task decomposition; fast and cost-efficient
Common node	gpt-4o-mini	General questions; speed over capability
All Summarization nodes	gpt-4o-mini	Cost-efficient; runs frequently on every agent cycle

Strategy: gpt-4.1 for security-critical and response-quality paths; gpt-4o-mini for high-frequency, cost-sensitive nodes

Architecture Summary

Security First

- Gatekeeper on every query
- Prompt injection detection
- Security threat blocking
- gpt-4.1 on critical paths

Smart Routing

- Query classification
- History-aware answering
- Multi-agent task decomposition
- Domain-specific delegation

Resource Control

- Token budgets at every level
- Step limits prevent loops
- Tool response summarization
- Per-cluster usage quotas

Cost Efficiency

- gpt-4o-mini for high-frequency nodes
- RAG for doc retrieval
- Pagination limits on K8s API
- Rolling summarization